MILITARY SPECIFICATION

PRELIMINARY PLANS AND TEST PROCEDURE REQUIREMENTS

FOR RADIAC INSTRUMENTS

1. SCOPE

1.1 This specification covers the requirements for preliminary plans and test procedures to be furnished during the procurement of radiac instruments (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 This section is not applicable to this specification.

3. REQUIREMENTS

3.1 Preliminary plans.- Preliminary plans shall be furnished for all equipments unless specifically waived by the bureau or agency concerned, and shall be in accordance with the contractor's commercial procedures. Plans shall be multi-detail, and content shall be in accordance with 3.1.7.

3.1.1 Information required shall be presented on four preliminery plans as follows:

(a) Assembly views.
(b) Electrical schematic diagram.
(c) Wiring diagram.
(d) Block diagram.

3.1.2 The number of reference plans and the amount of detail shall be held to a minimum consistent with fulfillment of their funtion.

3.1.3 Where an instrument consists of two or more assemblies separate preliminary plans may be prepared to cover one or more of the assemblies.

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3.1.4 Plan size shall be in accordance with the contractor's commercial procedures. Where multi-sheet plans are furnished, each sheet shall bear the same plan number, the sheet number, and the total number or sheets.

3.1.5 Preliminary plans shall be clear and legible.

3.1.6 Block diagram. - A block diagram shall be furnished with the preliminary plans. The diagram shall clearly depict the major functions of the equipment.

3.1.7 Contents of preliminary plans. - Preliminary plans shall include the following minimum information as applicable:

(a) Descriptive data for the equipment.
  (1) Type of equipment.
  (2) Applicable environmental conditions.
  (3) Degree of enclosure.
  (4) Temperature ranges (ambient).
  (5) Shock and vibration requirements.
  (6) Range or ratings (including accuracies) as applicable.
  (7) Power requirements; voltages, phase, frequency, watts or voltamperes, power factor, allowable variations in voltage and frequency.
  (8) Heat dissipation.
  (9) Manufacturer's designation.
  (10) Mounting.
(b) A statement that the equipment is in accordance with the particular referenced individual equipment specification. Where there are exceptions, the exceptions shall be listed.
(c) A list of reference drawings.
(d) List of material in accordance with figure 1. The list of material shall be on the same sheet as the assembly views required in (e).

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(e) Assembly views consisting of plan, elevation and section views as required to show clearly the details of the mechanical design, construction, and assembly of the equipment and to identify each part and its location. Each part shall be identified by its reference designation and by its piece number as assigned in the list of materials. The following minimum data shall be included on the assembly views:

(1) Overall dimensions.
(2) Weight of complete assembly (estimate).
(3) A front panel view (drawn to scale) showing all visible parts and markings.

(f) An electrical schematic diagram on a single sheet to represent clearly the operation of the equipment. Notes describing circuit operation, and arrows with accentuated lines indicating the path of signal flow, shall be employed when necessary to clarify the operation of the equipment. Signal flow shall be conventional current flow and not electron flow.

(1) A thin broken line shall be used to represent the boundaries of each assembly or subassembly which is intended for multiple use. All terminals with their numbers and markings shall be shown. Terminals for external connections shall also indicate the type of signal, power, and/or ground as appropriate. All unused terminals shall be shown and so marked.

(2) Each part, such as resistors, capacitors or relays, shall be given a unique reference designation consisting of a letter denoting the type of part and a number assigned consecutively. (The reference designation need not be changed where an assembly is intended for multiple use.) The numbers shall be assigned in a logical sequence of the electrical current or signal flow through the circuit.

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(3) The location of parts shall be such that the signal flow through the circuit is generally from left to right. The physical location of parts in the equipment shall not influence the location of the part in the schematic diagram. Sections of switches, relay coils and contacts, and individual terminals of terminal boards or connectors may be shown in separate locations for clarity; however, the basic part reference designation shall be retained.

(g) A wiring diagram which includes the following information:

(1) All parts in correct relation to physical location. Terminals shall be clearly shown and marked and all wiring between parts shown.

(2) Appropriate notes shall designate wire size and type.

(3) Reference designation of parts shall conform to those assigned in the schematic diagram.

3.1.8 Approval procedures.- All preliminary plans shall be approved by the bureau or agency concerned. Any work (such as procurement of material or fabrication of any part of the equipment) done by the contractor in advance of approval of preliminary plans is at the contractor's own risk.

3.1.9 After preliminary plans have been approved, no changes in design or construction shall be made without the approval of the bureau or agency concerned, except that correction of errors and omissions shall be made with the approval of the Government inspector. Three copies of each such corrected plan shall be immediately forwarded to the bureau or agency concerned.

3.1.10 Microfilm requirements.- Submission of preliminary plans to the bureau or agency concerned does not relieve the manufacturer of any contract requirement for submission of microfilm of engineering drawings. Microfilm of drawings shall be as specified in the contract or order.

3.2 Test procedures.- Proposed detailed test procedures for the tests specified in the individual equipment specifications shall be prepared by the equipment manufacturer. The test procedures shall indicate where the tests are to be conducted, the sequence in which they are to be conducted, detailed step by step method of test and test equipment requirements. They shall also indicate whether test facilities and equipment are currently available or anticipated availability when not currently available. The test procedure format shall contain provisions for recording test results in such a manner as to permit recording of each required test in a clear and concise form.

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3.2.1 Approval. - The test procedures shall be submitted to the bureau or agency concerned for approval. Approval of the test procedures shall be obtained prior to setting up test equipment for preproduction or quality conformance inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 This section is not applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Preliminary plans. - The preliminary plans required herein are primarily intended to:

(a) Provide design information to assure conformance to requirements of the applicable equipment specification.
(b) Evaluate the suitability of the design for the intended Naval use.
(c) Evaluate performance and maintenance capability.

Preparing activity:
Navy - Ships
(Project MISC-N181Sh)

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List of materials/equipment

PC Number	Number Required	Name of Part	Material or MIL part no.	Material spec. or part spec.	Contractor		Manufacturer		Spec. Sheet	Finish	Remarks
					Drawing number	Service part no.	Name	Service part or drawing no.			

1. Width of columns may be varied to suit the data to be entered therein.
2. The two columns for the manufacturer's name and service part number may be omitted if the contractor is the manufacturer of all items.
3. Where the contractor's drawing numbers and service part numbers are identical, the two columns for this information may be made a single column headed "Contract or Drawing No. and Service Part No.".
4. The letters "QPL" may be entered in the "Manufacturer's Name" column for resistors, capacitors, fuses, silicon rectifiers, and transistors if the specification referenced specifically delineates the exact physical and electrical characteristics for the part, the part is purchased from a supplier on the qualified product list, the part is inspected per the referenced part specification.
5. When a matched subassemblies or matched or selected parts are used, the requirements for matching or selection shall be specified in a note.

Figure 1 - Sample list of material.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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